13.2 Gravitational Potential

Question Paper

Course	CIE A Level Physics
Section	13. Gravitational Fields
Topic	13.2 Gravitational Potential
Difficulty	Easy

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

State the physical quantity defined as *the energy possessed by an object due to its position in a gravitational field*.

[1 mark]

Question 1b

Define *gravitational potential* at a point.

[2 marks]

Question 1c

Gravitational potential, φ, is also defined by the equation

$$\varphi = -\frac{GM}{r}$$

(i)

Identify each quantity in the equation.

[3]

(ii)

State the significance of the negative sign in the equation.

[2]

[5 marks]

Question 1d

Fig. 1.1 shows a person on the surface of the Earth, a ball falling towards the centre of the Earth and a satellite in orbit around the Earth.

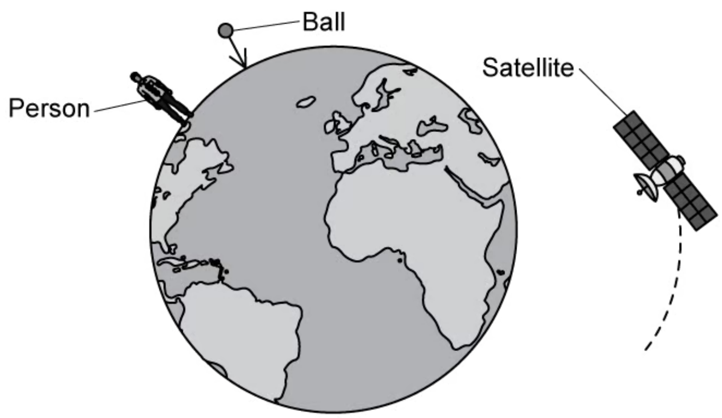

Fig. 1.1

Identify the object which experiences the largest gravitational potential.

[1 mark]

Question 2a

Place a tick (✓) next to the equation in Table 1.1 which represents the gravitational potential energy of two point masses *m* and *M* separated by a distance *r*.

Table 1.1

Equation	
$\varphi = -\dfrac{GM}{r}$	
$E_p = -\dfrac{GMm}{r}$	
$E_p = mgh$	
$g = \dfrac{GM}{r^2}$	

[1 mark]

Question 2b

The diagram in Fig. 1.2 shows two satellites m_1 and m_2 in orbit around the moon.

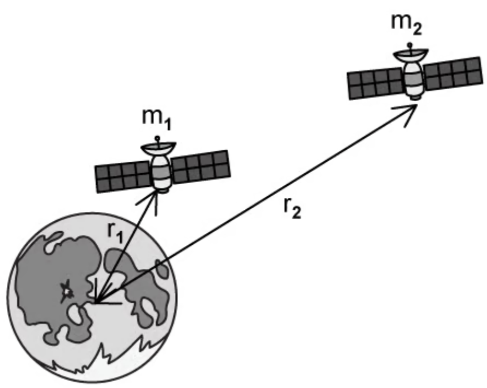

Fig. 1.2

Satellite m_1 is in orbit at a distance of r_1 from the centre of the moon and satellite m_2 is in orbit at a distance of r_2 from the centre of the moon.

State which of the satellites has the higher gravitational potential energy.

[1 mark]

Question 2c

Satellite m_1 has a mass of 300 kg and orbits above the centre of the moon at a radius r_1 of 500 m.

The mass of the moon is 7×10^{22} kg.

Calculate the gravitational potential energy of the satellite in this position. Newton's gravitational constant, $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

[2 marks]

Question 2d

State why the equation for gravitational potential energy in **(a)** has a negative sign and why your answer to **(c)** is negative.

[3 marks]

Question 3a

Explain why gravitational potential is always negative.

[2 marks]

Question 3b

Fig. 1.1 shows a satellite orbiting Mars, M. The satellite is moves from orbit X to orbit Y.

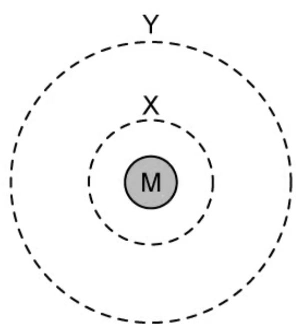

Fig. 1.1

The gravitational potential due to Mars in each of these orbits is:

Orbit X: -3.56 MJ kg^{-1}
Orbit Y: -1.23 MJ kg^{-1}

Calculate the gravitational potential difference as the satellite moves from orbit X to orbit Y.

[3 marks]

Question 3c

The gravitational potential energy at a particular point is given by the equation:

$$E_p = -\frac{GMm}{r}$$

Define each of the terms in the equation.

[4 marks]

Question 3d

A satellite is in orbit at 2×10^6 m above the surface of the Earth. The Earth has a radius of 6.4×10^6 m.

Calculate the orbital radius of the satellite.

[3 marks]

Model Answers: Easy

1a

(a) This is the definition for:

- Gravitational potential energy; [1 mark]

[Total: 1 mark]

Make sure you understand the difference between **gravitational potential** and **gravitational potential energy**

1b

(b) Gravitational potential is defined as:

- The work done per unit mass; [1 mark]
- In bringing a test mass from infinity (to the point); [1 mark]

[Total: 2 marks]

1c

(c)

(i) The quantities in the equation are as follows:

- G = Newton's gravitational constant (6.67×10^{-11} N m^2 kg^{-2}); [1 mark]
- M = mass of the body (producing the gravitational field); [1 mark]
- r = distance from the centre of mass to the point mass; [1 mark]

(ii) The negative sign shows that:

- The gravitational force is (always) attractive; [1 mark]
- As r decreases, M does work **OR** work is done by the masses as they come together; [1 mark]

[Total: 5 marks]

1d

(d) The object experiencing the largest gravitational potential is:

- The satellite; [1 mark]

[Total: 1 mark]

Consider the **equation** for gravitational potential:

$$\varphi \uparrow = -\frac{GM}{r \downarrow}$$

As ***r* decreases *φ* increases**

2a

(a) The correct equation for gravitational potential energy is:

Equation	
$\varphi = -\frac{GM}{r}$	
$E_p = -\frac{GMm}{r}$	✓
$E_p = mgh$	
$g = \frac{GM}{r^2}$	

- Tick placed next to $E_p = -\frac{GMm}{r}$ [1 mark]

[Total: 1 mark]

2b

(b) The satellite with the highest gravitational potential energy is:

- Satellite m_2; [1 mark]

[Total: 1 mark]

Objects **further away** from the **surface** of the Earth have a **greater gravitational potential** because they have the potential to **fall further** and **gain more energy** as they move towards the Earth.

2c

(c) Calculate the gravitational potential energy of the satellite:

List the known quantities:

- Mass of satellite, $m_1 = 300$ kg
- Radius of orbit, $r_1 = 500$ m
- Mass of moon, $M = 7 \times 10^{22}$ kg
- Newton's gravitational constant, $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

Substitute the known quantities into the equation from part (a):

- $E_p = -\frac{GMm_1}{r_1}$
- $E_p = -\frac{(6.67 \times 10^{-11}) \times (7 \times 10^{22}) \times (300)}{(500)}$ [1 mark]
- $E_p = -2.8 \times 10^{12}$ J [1 mark]

[Total: 2 marks]

2d

(d) Values for gravitational potential energy are negative because:

- Gravitational potential at infinity is defined as being zero; [1 mark]
- Gravitational forces are always attractive; [1 mark]
- So potential energy decreases as the masses are brought closer together; [1 mark]

[Total: 3 marks]

3a

(a) Gravitational potential is always negative because:

- Gravitational potential is equal to 0 at infinity; [1 mark]

Any **one** from:

- (The gravitational force is attractive) so work must be done (on a mass) to reach infinity; [1 mark]
- Potential energy decreases as a body moves from infinity towards the Earth; [1 mark]

[Total: 2 marks]

3b

(b) Calculate the change in gravitational potential as the satellite moves from orbit X to orbit Y:

List the known quantities:

- Potential of X, φ_X = –3.56 MJ kg^{-1}
- Potential of Y, φ_Y = –1.23 MJ kg^{-1}

Calculate the gravitational potential difference between orbits X and Y:

- $\Delta\varphi = \varphi_Y - \varphi_X$ [1 mark]
- $\Delta\varphi = (-1.23) - (-3.56)$ [1 mark]
- $\Delta\varphi = 2.33 \text{ MJ kg}^{-1}$ [1 mark]

[Total: 3 marks]

3c

(c) Definitions of each of the terms in the equation are as follows:

- G = the gravitational field constant; [1 mark]
- M = the mass of the body producing the gravitational field; [1 mark]
- m = the mass of the object in the gravitational field; [1 mark]
- r = the distance from the centre of mass to the centre of the point mass; [1 mark]

[Total: 4 marks]

3d

(d) Calculate the orbital radius of the satellite:

- Height of orbit above Earth's surface, $h = 2 \times 10^6$ m
- Radius of Earth, $r = 6.4 \times 10^6$ m

Calculate the orbital radius of the satellite, *R*:

- $R = r + h$ [1 mark]
- $R = (6.4 \times 10^6) + (2 \times 10^6)$ [1 mark]
- $R = 8.4 \times 10^6$ m [1 mark]

[Total: 3 marks]

Remember that the **orbital radius** of a satellite is positioned from the **centre** of the object with the gravitational field.